SECTION 1

MNIST

- **Dataset role:** MNIST's 70k 28×28 grayscale images (784 features per row) is the customary benchmark for new classifiers.
- **Labels and splits:** Targets are cast to integers; the canonical split keeps 60k train / 10k test, with the test set reserved until final evaluation.
- **Exploration caveat:** Visualizing digits is fine on the training portion, but peeking at the test set biases later choices.
- **Complexity:** Wide stylistic variation among writers shows why linear models need careful preprocessing to compete.

SECTION 2

Training a Binary Classifier

- **5-detector task:** A boolean target (`y == 5`) turns multiclass data into binary classification for simpler metric study.
- **SGD classifier:** `SGDClassifier` scales to large sparse data and supports online-style updates with reproducibility via `random_state`.
- **Prediction flow:** After `fit`, `predict` outputs whether the model believes the instance is the positive class.
- **Evaluation need:** High-level accuracy on cross-validation still requires deeper metrics before trusting deployment.

SECTION 3

Performance Measures

- **Accuracy pitfall:** A majority-class "never 5" classifier scores ~90% accuracy while learning nothing about fives—skewed priors inflate accuracy.
- **Confusion matrix:** Counts of true/false positives and negatives show where errors concentrate.
- **Precision, recall, F1:** Precision is trust in positive calls; recall is coverage of actual positives; F1 harmonizes both when a single scalar helps comparison.
- **Threshold trade-off:** `decision_function` scores versus a cutoff move along a precision–recall frontier; `precision_recall_curve` and ROC analysis guide cut selection.
- **ROC and AUC:** Plotting TPR vs. FPR summarizes ranking quality; PR curves are preferred when positives are rare.

SECTION 4

Multiclass Classification

- **Native vs. meta strategies:** Some algorithms handle all ten digits directly; strict binary models use one-vs-rest (OvR) or one-vs-one (OvO) decompositions.
- **SVM behavior:** `SVC` on MNIST may train many binary machines and pick the class with the strongest vote or score.
- **Scaling lift:** `StandardScaler` on pixel features often improves linear multiclass SGD accuracy noticeably.
- **Wrappers:** `OneVsRestClassifier` forces OvR when the base estimator is inherently binary.

SECTION 5

Error Analysis

- **Confusion matrix plots:** Normalizing rows highlights which true digits get mistaken for which labels (e.g., 3 vs. 5, 8 confusion).
- **Remediation ideas:** Targeted data collection, feature engineering, or preprocessing (centering, deskewing) address systematic confusions.
- **Qualitative review:** Side-by-side misclassified images reveal ambiguous glyphs versus model limitations (linear decision surfaces).

SECTION 6

Multilabel Classification

- **Multiple binary tags:** Each instance may activate several independent labels (e.g., large digit and odd digit simultaneously).
- **KNeighbors example:** Some estimators accept two-dimensional binary targets and predict multiple outputs per row.
- **Metrics:** Per-label F1 scores aggregate with `average="macro"` or `"weighted"` depending on whether rare labels should dominate.

SECTION 7

Multioutput Classification

- **Per-pixel targets:** Denoising MNIST treats each intensity as a label, blending classification-shaped outputs with regression-like values.
- **Setup:** Noisy inputs map to clean training images; a neighbor classifier can predict entire flattened vectors.
- **Scope:** Multioutput extends multilabel by allowing multiclass or continuous targets per output slot.